

Climate 5000 Series - Gen 4

36K BTU Single Zone Regular Light Commercial System

4-Way Cassette IDU

Engineering
Submittal
Sheet



BOSCH

Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

36K Btu Single Zone Regular Light Commercial System - 4-Way Cassette IDU

Indoor part number	8-733-962-682
Indoor model number	BMS500-AAU036-1AHCXD
Outdoor part number	8-733-962-692
Outdoor model number	BMS500-AAS036-1CSXLD

Efficiency Data AHRI 210/240 2023

SEER2	20.0
EER2	9.6
HSPF2 Region IV	10.3
HSPF2 Region V	7.9
COP(47F)	3.31
COP (5F)	2.20

Performance Data AHRI 210/240 2023

Cooling	Capacity Range	6400~39800
	Rated Capacity	36000
	Current	16.4
	Input Power	3720
Heating	Capacity Range	14400~42200
	Rated Capacity (47F)	38000
	Rated Capacity (5F)	26600
	Current	15.0
	Input Power	3365

Electrical Data

Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)	3.0 / 33.0
Max. fuse indoor/outdoor (A)	15.0 / 35.0

Compressor Data

Type:	Rotary
Refrigerant oil /oil charge (ml)	VG74 / 1000
Input (W)	2765
Rated current (RLA/A)	5.4

Piping

Refrigerant precharge (oz/kg)	91.71 / 2.60
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	3/8" / Ø9.52
Gas side (in/mm)	3/4" / Ø19.0
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.32 / 30
Max. refrigerant pipe length (ft/m)	246.00 / 75
Max. difference in height (ft/m)	98.40 / 30

Operation Range

ODU	Cooling (°F / °C)	-13~122 / -25~50
	Heating (°F / °C)	-13~75 / -25~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		8.94
IDU	Fan motor model	ZKFN-125-8-1
	Fan motor qty (ea)	1
	Fan motor input (W)	200.0
	Fan speed (RPM)	696 / 632 / 568
	Air flow, Turbo / H / M / L / Si (CFM)	1041.8 / 947.6 / 853.5 / 753.4 / 659.2
	Approx. Air Throw Range ft (m)	13.1 / 4
	Noise level, Turbo / H / M / L / Si (dB(A))	54 / 52.5 / 47.0 / 39.5 / 36.5
	Net weight (lbs / kg)	59.30 / 26.9
	Gross weight (lbs / kg)	71.43 / 32.4
	Panel net weight (lbs / kg)	13.23 / 6
ODU	Gross panel weight (lbs / kg)	19.84 / 9
	Drain pipe diameter (mm)	ODΦ25mm
	Fan motor model	ZKFN-120-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	170.0
	Fan motor speed (RPM)	950 / 800 / 500
	Air flow, Max (CFM)	2413.3
	Noise level, Max (dB(A))	63.0
	Net weight (lbs / kg)	152.34 / 69.1
	Gross weight (lbs / kg)	162.48 / 73.7

Accessories

BMS-RCG-XXD (8733962703)	Decorative Grille - 24K ~ 48K
BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface

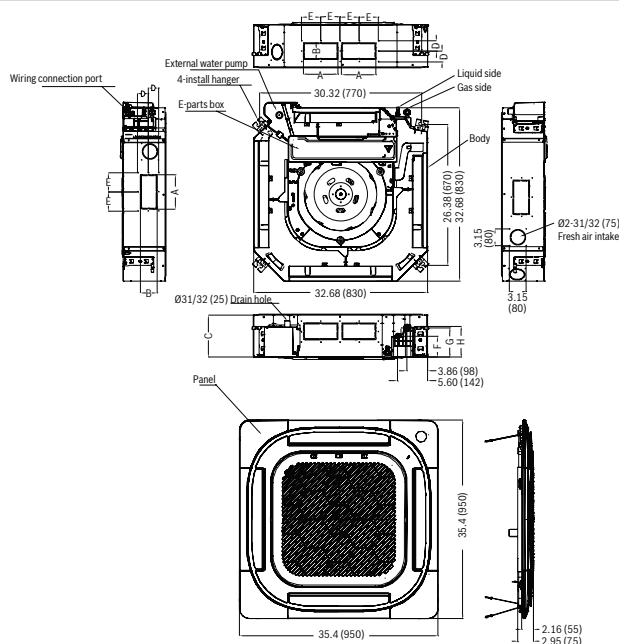
Standard Parts

8-733-969-065	Remote Control
---------------	----------------

36K Btu Single Zone Regular Light Commercial System - 4-Way Cassette IDU

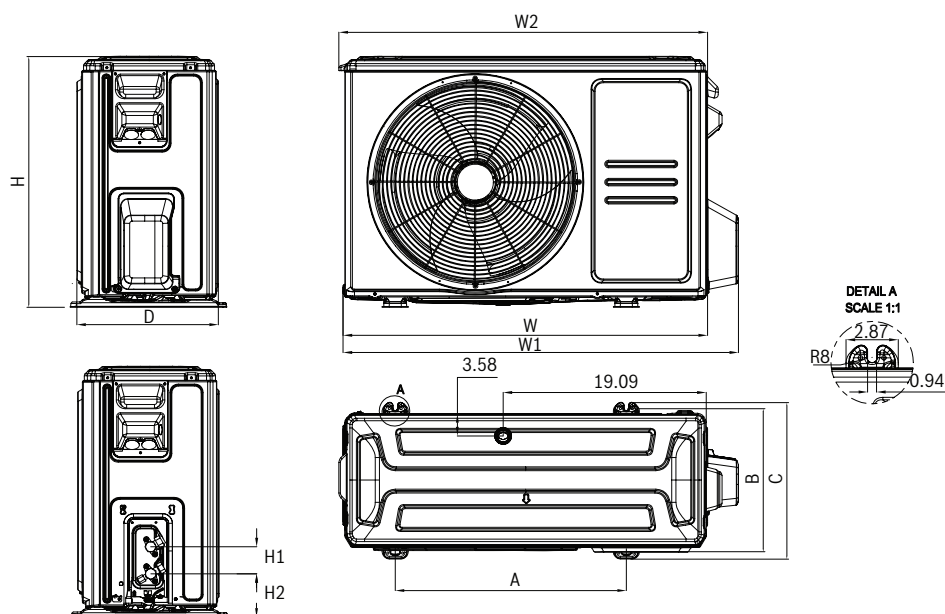


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU036-1AHCXD	32.68	32.68	9.65

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)	H1 (in)	H2 (in)
BMS500-AAS036-1CSXLD	37.24	16.14	31.89	40.55	37.24	26.5	15.87	17.9	2.36	4.25

Climate 5000 Series - Gen 4

36K Btu Single Zone Regular Light Commercial System - 4-Way Cassette IDU



BOSCH

Cooling Performance - Indoor Dry Bulb Temperature at 1041 CFM																						
ODU DB(°F)	IDU WB (°F)	60.8°F (16.0°C)					64.4°F (18.0°C)					67°F (19.4°C)					71.6°F (22.0°C)					
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	
-22°F (-30.0°C)	TC S/T PI	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	
-13°F (-25.0°C)	TC S/T PI	35011 0.61 1.90	35058 0.72 1.90	35096 0.82 1.90	35097 0.82 1.90	35144 0.93 1.90	37121 0.58 2.01	37165 0.68 2.01	37201 0.77 2.01	37209 0.77 2.01	37253 0.88 2.02	39338 0.54 2.13	39389 0.64 2.13	39433 0.73 2.13	39441 0.73 2.14	39485 0.83 2.14	41709 0.51 2.26	41754 0.60 2.26	41799 0.68 2.26	41806 0.68 2.26	41858 0.78 2.27	
-4°F (-20.0°C)	TC S/T PI	37704 0.62 1.96	37755 0.73 1.97	37795 0.83 1.97	37797 0.83 1.97	37848 0.94 1.97	39976 0.59 2.08	40023 0.69 2.08	40063 0.78 2.09	40071 0.78 2.09	40118 0.89 2.09	42364 0.55 2.21	42419 0.65 2.21	42467 0.74 2.21	42475 0.74 2.21	42522 0.84 2.21	44917 0.52 2.34	44966 0.61 2.34	45015 0.70 2.34	45022 0.70 2.34	45078 0.79 2.35	
0°F (-17.8°C)	TC S/T PI	38901 0.63 1.99	38953 0.73 2.00	38995 0.84 2.00	38997 0.84 2.00	39049 0.95 2.00	41245 0.59 2.11	41294 0.69 2.12	41335 0.79 2.12	41343 0.79 2.12	41392 0.90 2.12	43709 0.56 2.24	43766 0.65 2.24	43815 0.74 2.24	43823 0.74 2.25	43872 0.85 2.25	46343 0.53 2.37	46393 0.62 2.38	46444 0.70 2.38	46451 0.70 2.38	46509 0.80 2.38	
5°F (-15.0°C)	TC S/T PI	40397 0.63 2.03	40451 0.74 2.03	40495 0.84 2.04	40497 0.84 2.04	40551 0.96 2.04	42831 0.60 2.15	42882 0.70 2.16	42925 0.80 2.16	42933 0.80 2.16	42984 0.91 2.16	45390 0.56 2.28	45449 0.66 2.28	45500 0.75 2.29	45508 0.75 2.29	45559 0.86 2.29	48125 0.53 2.42	48178 0.62 2.42	48230 0.71 2.42	48237 0.71 2.42	48298 0.81 2.43	
14°F (-10.0°C)	TC S/T PI	39353 0.62 2.25	39406 0.73 2.26	39449 0.83 2.26	39451 0.83 2.26	39504 0.94 2.26	41725 0.59 2.39	41774 0.69 2.39	41816 0.78 2.39	41824 0.78 2.40	41874 0.89 2.40	44218 0.55 2.53	44275 0.65 2.54	44325 0.74 2.54	44333 0.74 2.54	44382 0.84 2.54	46882 0.52 2.69	46933 0.61 2.69	46984 0.69 2.69	46991 0.69 2.69	47050 0.79 2.69	
17°F (-8.3°C)	TC S/T PI	39006 0.62 2.33	39058 0.72 2.33	39100 0.82 2.33	39102 0.82 2.33	39155 0.94 2.34	41356 0.58 2.47	41405 0.68 2.47	41446 0.78 2.47	41454 0.78 2.47	41504 0.88 2.48	43827 0.55 2.62	43884 0.64 2.62	43933 0.73 2.62	43941 0.73 2.62	43990 0.83 2.63	46468 0.52 2.77	46518 0.61 2.78	46569 0.69 2.78	46576 0.69 2.78	46634 0.79 2.78	
23°F (-5.0°C)	TC S/T PI	38352 0.61 2.40	38404 0.72 2.41	38445 0.81 2.41	38447 0.81 2.41	38499 0.93 2.41	40664 0.58 2.55	40712 0.68 2.55	40752 0.77 2.55	40760 0.77 2.55	40808 0.88 2.56	43093 0.54 2.70	43149 0.64 2.70	43197 0.72 2.71	43205 0.72 2.71	43253 0.83 2.71	45689 0.51 2.86	45739 0.60 2.87	45789 0.68 2.87	45796 0.68 2.87	45853 0.78 2.87	
35°F (1.7°C)	TC S/T PI	37046 0.60 2.55	37095 0.70 2.56	37135 0.80 2.56	37137 0.80 2.56	37187 0.91 2.56	39278 0.56 2.71	39325 0.66 2.71	39364 0.75 2.71	39371 0.75 2.72	39418 0.86 2.72	41625 0.53 2.87	41679 0.62 2.87	41725 0.71 2.88	41733 0.71 2.88	41780 0.81 2.88	44133 0.50 3.04	44181 0.59 3.05	44229 0.67 3.05	44236 0.67 3.05	44291 0.76 3.05	
41°F (5.0°C)	TC S/T PI	36331 0.61 2.39	36380 0.72 2.39	36419 0.82 2.39	36420 0.82 2.39	36469 0.93 2.39	38520 0.58 2.53	38566 0.68 2.53	38604 0.77 2.54	38612 0.77 2.54	38657 0.88 2.54	40822 0.54 2.68	40875 0.64 2.68	40920 0.73 2.69	40928 0.73 2.69	40973 0.83 2.69	43281 0.51 2.84	43328 0.60 2.84	43375 0.69 2.85	43382 0.69 2.85	43436 0.78 2.85	
47°F (8.3°C)	TC S/T PI	35616 0.63 2.22	35664 0.73 2.22	35702 0.83 2.22	35704 0.83 2.22	35752 0.95 2.22	37762 0.59 2.35	37807 0.69 2.35	37844 0.79 2.36	37852 0.79 2.36	37897 0.90 2.36	40018 0.56 2.49	40070 0.65 2.49	40115 0.74 2.50	40122 0.74 2.50	40167 0.85 2.50	42429 0.53 2.64	42476 0.62 2.64	42522 0.70 2.65	42528 0.70 2.65	42581 0.80 2.65	
50°F (10.0°C)	TC S/T PI	35683 0.62 2.25	35731 0.73 2.26	35770 0.82 2.26	35771 0.82 2.26	35819 0.94 2.26	37834 0.58 2.39	37878 0.68 2.39	37916 0.78 2.39	37923 0.78 2.40	37968 0.89 2.40	40094 0.55 2.53	40146 0.65 2.53	40191 0.73 2.54	40198 0.73 2.54	40243 0.84 2.54	42510 0.52 2.68	42556 0.61 2.69	42602 0.69 2.69	42609 0.69 2.69	42662 0.79 2.69	
59°F (15.0°C)	TC S/T PI	35885 0.60 2.36	35933 0.70 2.37	35972 0.80 2.37	35974 0.80 2.37	36022 0.91 2.37	38048 0.56 2.50	38093 0.66 2.51	38130 0.75 2.51	38138 0.75 2.51	38183 0.86 2.51	40321 0.53 2.65	40373 0.62 2.66	40418 0.71 2.66	40418 0.71 2.66	40426 0.81 2.66	40471 0.81 2.66	42750 0.50 2.81	42797 0.59 2.82	42843 0.67 2.82	42850 0.67 2.82	42903 0.76 2.82
68°F (20.0°C)	TC S/T PI	36087 0.58 2.47	36135 0.68 2.48	36174 0.77 2.48	36176 0.77 2.48	36225 0.88 2.48	38262 0.54 2.62	38307 0.64 2.62	38345 0.72 2.63	38352 0.72 2.63	38398 0.83 2.63	40547 0.51 2.78	40600 0.60 2.78	40645 0.68 2.78	40653 0.68 2.78	40698 0.78 2.79	42991 0.48 2.95	43037 0.57 2.95	43084 0.65 2.95	43091 0.65 2.95	43144 0.74 2.96	
77°F (25.0°C)	TC S/T PI	36289 0.55 2.58	36337 0.65 2.59	36377 0.74 2.59	36378 0.74 2.59	36427 0.84 2.59	38476 0.52 2.74	38521 0.61 2.74	38559 0.70 2.74	38567 0.70 2.74	38613 0.80 2.75	40774 0.49 2.90	40827 0.58 2.90	40873 0.66 2.91	40880 0.66 2.91	40926 0.75 2.91	43231 0.47 3.08	43278 0.55 3.08	43325 0.62 3.08	43332 0.62 3.08	43386 0.71 3.09	
82°F (27.8°C)	TC S/T PI	36248 0.56 2.78	36296 0.66 2.79	36336 0.75 2.79	36337 0.75 2.79	36386 0.85 2.79	38432 0.53 2.95	38478 0.62 2.95	38516 0.70 2.96	38523 0.70 2.96	38569 0.80 2.96	40728 0.50 3.13	40781 0.58 3.13	40827 0.66 3.13	40834 0.66 3.13	40880 0.76 3.14	43182 0.47 3.31	43229 0.55 3.32	43276 0.63 3.32	43283 0.63 3.32	43337 0.71 3.33	
86°F (30.0°C)	TC S/T PI	36022 0.58 3.12	36070 0.69 3.12	36109 0.78 3.12	36111 0.78 3.12	36160 0.89 3.13	38193 0.55 3.30	38238 0.65 3.31	38276 0.74 3.31	38283 0.74 3.31	38329 0.84 3.32	40475 0.52 3.50	40527 0.61 3.51	40572 0.69 3.51	40580 0.69 3.51	40625 0.79 3.52	42913 0.49 3.71	42960 0.58 3.72	43007 0.65 3.72	43013 0.65 3.72	43067 0.75 3.73	
95°F (35.0°C)	TC S/T PI	35514 0.64 3.87	35562 0.75 3.88	35600 0.85 3.88	35601 0.85 3.88	35649 0.97 3.89	37654 0.60 4.10	37699 0.71 4.11	37736 0.81 4.11	37743 0.81 4.11	37788 0.92 4.12	39904 0.57 4.35	39955 0.67 4.36	40000 0.76 4.36	40007 0.76 4.36	40052 0.87 4.37	42308 0.54 4.61	42354 0.63 4.62	42400 0.72 4.62	42407 0.72 4.62	42459 0.82 4.63	
104°F (40.0°C)	TC S/T PI	28966 0.69 3.30	29005 0.81 3.30	29036 0.92 3.30	29037 0.92 3.30	29077 1.00 3.31	30712 0.65 3.50	30748 0.76 3.50	30778 0.87 3.50	30784 0.87 3.50	30821 0.99 3.51	32546 0.61 3.70	32589 0.72 3.71	32625 0.82 3.71	32631 0.82 3.71	32667 0.93 3.72	34507 0.58 3.93	34545 0.68 3.93	34583 0.77 3.94	34588 0.77 3.94	34631 0.88 3.94	
110°F (43.0°C)	TC S/T PI	25037 0.72 2.95	25071 0.84 2.96	25098 0.96 2.96	25099 0.96 2.96	25133 1.00 2.96	26546 0.68 3.13	26578 0.79 3.13	26604 0.90 3.14	26609 0.90 3.14	26641 1.00 3.14	28132 0.64 3.32	28169 0.75 3.32	28200 0.85 3.32	28205 0.85 3.33	28237 0.97 3.33	29827 0.60 3.52	29860 0.71 3.52	29892 0.80 3.52	29897 0.80 3.52	29934 0.91 3.53	
115°F (46.0°C)	TC S/T PI	20640 0.74 2.61	20668 0.87 2.62	20690 0.99 2.62	20691 0.99 2.62	20719 1.00 2.62	21884 0.70 2.77	21910 0.82 2.78	21931 0.93 2.78	21936 0.93 2.78	21962 1.00 2.78	23191 0.66 2.94	23221 0.77 2.94	23247 0.88 2.94	23252 0.88 2.95	23278 1.00 2.95	24589 0.62 3.11	24615 0.73 3.12	24642 0.83 3.12	24646 0.83 3.12	24677 0.95 3.13	
122°F (50.0°C)	TC S/T PI	14777 0.75 2.17	14797 0.88 2.17	14813 1.00 2.17	14814 1.00 2.17	14834 1.00 2.17	15668 0.73 2.30	15686 0.86 2.30	15702 0.98 2.30	15705 0.98 2.30	15723 1.00 2.30	16604 0.69 2.43	16625 0.81 2.44	16644 0.92 2.44	16647 0.92 2.44	16665 1.00 2.44	17604 0.65 2.58	17623 0.76 2.58	17642 0.87 2.59	17645 0.87 		

Climate 5000 Series - Gen 4

36K Btu Single Zone Regular Light Commercial System - 4-Way Cassette IDU



BOSCH

Heating Performance - Indoor Dry Bulb Temperature at 1041 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	/	/	/	/	/	/	/	/	/	/
-13°F(-25.0°C)	16921	16583	16500	16434	16286	3.13	3.15	3.16	3.17	3.18
-4°F(-20.0°C)	21331	20905	20800	20717	20530	3.37	3.40	3.41	3.41	3.43
0°F(-17.8°C)	23975	23495	23378	23284	23075	3.43	3.46	3.47	3.47	3.49
5°F(-15.0°C)	27279	26734	26600	26494	26255	3.51	3.54	3.54	3.55	3.57
14°F(-10.0°C)	28433	27864	27725	27614	27366	3.63	3.66	3.67	3.68	3.69
17°F(-8.3°C)	28818	28241	28100	27988	27736	3.67	3.70	3.71	3.72	3.73
22°F(-5.6°C)	29799	29203	29057	28940	28680	3.59	3.62	3.63	3.64	3.65
27°F(-2.8°C)	30780	30164	30013	29893	29624	3.52	3.54	3.55	3.56	3.57
32°F(0.0°C)	31761	31126	30970	30846	30568	3.44	3.46	3.47	3.48	3.49
35°F(1.7°C)	32349	31702	31544	31418	31135	3.39	3.42	3.42	3.43	3.44
37°F(2.8°C)	34222	33538	33370	33236	32937	3.42	3.44	3.45	3.46	3.47
42°F(5.6°C)	38904	38126	37935	37783	37443	3.48	3.51	3.52	3.52	3.54
44.6°F(7.0°C)	40776	39961	39761	39602	39246	3.51	3.53	3.54	3.55	3.56
47°F(8.3°C)	43585	42714	42500	42330	41949	3.54	3.57	3.58	3.59	3.60
52°F(11.1°C)	43150	42287	42075	41907	41530	3.32	3.35	3.35	3.36	3.37
57°F(13.9°C)	42714	41860	41651	41484	41111	3.10	3.12	3.13	3.14	3.15
59°F(15.0°C)	42540	41689	41481	41315	40943	3.01	3.03	3.04	3.04	3.06
62°F(16.7°C)	42279	41433	41226	41061	40692	2.87	2.90	2.90	2.91	2.92
64.4°F(18.0°C)	42070	41229	41022	40858	40491	2.77	2.79	2.80	2.80	2.81
69.8°F(21.0°C)	41600	40768	40564	40402	40038	2.53	2.55	2.55	2.56	2.57
75.2°F(24.0°C)	41130	40307	40105	39945	39585	2.28	2.30	2.31	2.31	2.32

Bolded values were tested. All other values are simulated and for reference only.